

CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 10 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

- Q1.** A body has a positive charge of 8×10^{-19} C. How many electrons were removed from this body to achieve this charge? (a) 2 (b) 5 (c) 8 (d) 10 [Chapter 1: Electric Charges & Fields | Confidence: Very High]
- Q2.** A test charge q_0 is moved along the equatorial line of an electric dipole from a distance r_1 to a distance r_2 . The work done by the electrostatic force in this process is: (a) Positive and non-zero (b) Negative and non-zero (c) Exactly zero (d) Dependent on the magnitude of the dipole moment [Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]
- Q3.** The drift velocity (v_d) of free electrons in a metallic conductor subjected to an electric field E varies as: (a) $v_d \propto E^2$ (b) $v_d \propto \sqrt{E}$ (c) $v_d \propto E$ (d) $v_d \propto 1/E$ [Chapter 3: Current Electricity | Confidence: Very High]
- Q4.** The magnetic field at the centre of a current-carrying circular loop of radius r carrying current I is B . If the current is doubled and the radius is perfectly halved, the new magnetic field at the centre will be: (a) B (b) $2B$ (c) $4B$ (d) $8B$ [Chapter 4: Moving Charges & Magnetism | Confidence: High]
- Q5.** Above its Curie temperature, a ferromagnetic material fundamentally behaves as a: (a) Diamagnetic material (b) Paramagnetic material (c) Superconductor (d) Non-magnetic material [Chapter 5: Magnetism & Matter | Confidence: Very High]
- Q6.** The fundamental physical principle behind Lenz's Law of electromagnetic induction is the law of conservation of: (a) Electric Charge (b) Linear Momentum (c) Energy (d) Angular Momentum [Chapter 6: Electromagnetic Induction | Confidence: Very High]
- Q7.** The electrical power factor of a purely inductive AC circuit is exactly: (a) 1 (b) 0.5 (c) 0 (d) Infinity [Chapter 7: Alternating Current | Confidence: Very High]
- Q8.** Among the following electromagnetic waves, which one possesses the maximum penetrating power through matter? (a) Microwaves (b) Ultraviolet rays (c) X-rays (d) Gamma rays [Chapter 8: Electromagnetic Waves | Confidence: Very High]
- Q9.** A symmetric biconvex glass lens has an optical power of $+P$. If it is cut precisely into two identical symmetric halves along its principal axis, the optical power of each half will be: (a) P (b) $P/2$ (c) $2P$ (d) $P/4$ [Chapter 9: Ray Optics | Confidence: High]
- Q10.** In a standard Young's Double Slit Experiment (YDSE), if the monochromatic light source is completely replaced by a white light source, the central fringe on the observation screen will be: (a) Perfectly dark (b) Purely red (c) Purely blue (d) White [Chapter 10: Wave Optics | Confidence: Very High]
- Q11.** In the context of the photoelectric effect, if the intensity of the incident radiation is strictly increased while maintaining a constant frequency above the threshold, the stopping potential will:

(a) Increase linearly (b) Decrease linearly (c) Remain entirely unchanged (d) Drop to exactly zero
[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]

Q12. An electric dipole placed in a uniform external electric field will categorically experience: (a) A net force but strictly zero torque (b) A net torque but strictly zero net force (c) Both a net force and a net torque (d) Neither a net force nor a net torque [Chapter 1: Electric Charges & Fields | Confidence: Very High]

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false.

Q13. Assertion (A): The electric field lines are always directed strictly perpendicular to an equipotential surface. **Reason (R):** The work done by the electrostatic force in moving a charge between any two points on an equipotential surface is exactly zero. [Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]

Q14. Assertion (A): A charged particle projected perfectly parallel to a uniform magnetic field experiences strictly zero magnetic force. **Reason (R):** The magnetic Lorentz force on a charged particle is mathematically given by $\vec{F} = q(\vec{v} \times \vec{B})$, which becomes zero when the angle between $\vec{v}$ and $\vec{B}$ is zero. [Chapter 4: Moving Charges & Magnetism | Confidence: Very High]

Q15. Assertion (A): An artificial satellite with a metallic surface orbiting the Earth exactly in its equatorial plane will not have any motional emf induced across it. **Reason (R):** The vertical (perpendicular) component of the Earth's magnetic field is essentially zero precisely at the magnetic equator. [Chapter 6: Electromagnetic Induction | Confidence: High]

Q16. Assertion (A): The maximum kinetic energy of emitted photoelectrons strictly increases if the physical intensity of the incident monochromatic light is increased. **Reason (R):** Increasing the macroscopic intensity of light directly increases the individual energy possessed by each photon. [Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]

Section B: 2-Mark Questions (Very Short Answer)

Q17. A parallel plate air capacitor has a measured capacitance of 8 pF . What will be its new capacitance if the distance between its plates is strictly halved, and the entire space between them is filled with a dielectric substance of dielectric constant $K = 5$? [Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]

Q18. A battery of emf $\mathcal{E}$ and internal resistance r is connected across a variable external resistor. A steady current I flows. Write the mathematical relation between $\mathcal{E}$, V (terminal voltage), I , and r . If a cell of emf 1.5 V and internal resistance 0.5Ω is abruptly short-circuited, calculate the exact current drawn from it. [Chapter 3: Current Electricity | Confidence: High]

Q19. A circular conducting coil consisting of 50 turns and possessing a radius of 0.10 m carries a steady current of 2.0 A . Calculate the exact magnitude of the magnetic field generated strictly at its geometrical centre. (Take $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$). [Chapter 4: Moving Charges & Magnetism | Confidence: Very High]

Q20. State any two critical real-world applications of (i) X-rays and (ii) Gamma rays. [Chapter 8: Electromagnetic Waves | Confidence: Very High]

Q21. The threshold work function of a specific metal surface is exactly **3.0 eV**. Will photoelectric emission take place if a laser beam of wavelength **500 nm** is incident on it? Justify your answer mathematically. (Take $hc \approx 1240 \text{ eV nm}$). [Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]

Section C: 3-Mark Questions (Short Answer)

Q22. Deduce the mathematical expression for the macroscopic drift velocity (v_d) of free electrons in a metallic conductor subjected to an external electric field (E), in terms of the electronic relaxation time (τ). Hence, arrive at the microscopic form of Ohm's law ($J = \sigma E$). [Chapter 3: Current Electricity | Confidence: Very High]

Q23. Using Ampere's circuital law, systematically derive the expression for the magnetic field intensity ($\vec{B}$) strictly deep inside a long, straight, tightly wound current-carrying solenoid having n turns per unit length. [Chapter 4: Moving Charges & Magnetism | Confidence: Very High]

Q24. A rectangular conducting coil of area A and possessing N turns is rotated with a uniform angular velocity ω about an axis perpendicular to a uniform magnetic field B . Analytically derive the expression for the instantaneous alternating emf (ϵ) induced in the coil. [Chapter 6: Electromagnetic Induction | Confidence: High]

Q25. In a series LCR alternating current circuit, define the term 'Electrical Resonance'. Find the strict mathematical expression for the resonant frequency (f_r). What is the exact value of the power factor of this circuit at the condition of resonance? [Chapter 7: Alternating Current | Confidence: Very High]

Q26. A compound microscope consists of an objective lens of focal length **1.0 cm** and an eyepiece of focal length **5.0 cm**. The physical distance between the two lenses (tube length) is exactly **15 cm**. Calculate the location of the object required to form the final image strictly at infinity (normal adjustment). [Chapter 9: Ray Optics | Confidence: High]

Q27. In Young's double-slit experiment, the separation between the two coherent slits is **0.2 mm** and the observation screen is placed **1.0 m** away. It is observed that the **4th** bright fringe forms at a linear distance of **1.2 cm** from the central maximum. Calculate the exact wavelength of the monochromatic light used in the experiment. [Chapter 10: Wave Optics | Confidence: Very High]

Q28. (A) Carefully draw the equipotential surfaces created by a system of two equal and opposite point charges (an electric dipole). (B) Show mathematically that the electric potential at any point precisely on the equatorial plane of an electric dipole is exactly zero. [Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: Dielectrics in Capacitors When a dielectric material is introduced between the metallic plates of a capacitor, it becomes polarized under the influence of the external electric field. This polarization effectively reduces the net electric field inside the dielectric, thereby decreasing

the potential difference between the plates (if isolated) and invariably increasing the overall capacitance of the system. The ratio of the new capacitance to the original vacuum capacitance is termed the dielectric constant (K) of the material. Dielectrics are broadly classified into polar (having permanent molecular dipole moments) and non-polar types. (A) Give one common example of a polar molecule. (1 Mark) (B) What exactly happens to the net capacitance of a parallel plate capacitor if a thick metallic (conducting) slab of thickness equal to the plate separation d is inserted between its plates? (1 Mark) (C) A parallel plate capacitor has an initial capacitance C_0 and is connected to a battery of voltage V_0 . A dielectric slab of constant $K = 4$ is then introduced to completely fill the space between the plates, while the battery remains continuously connected. How does the total electrostatic energy stored in the capacitor quantitatively change? Explain. (2 Marks) [Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]

Q30. Case Study: Optical Fibers and Total Internal Reflection Optical fibers represent a revolutionary application of the phenomenon of total internal reflection (TIR). These thin, flexible strands of high-quality glass or plastic consist of an inner 'core' surrounded by an outer 'cladding'. When a light signal enters the core at a suitable angle, it undergoes repeated total internal reflections at the core-cladding boundary, allowing the signal to travel tremendous distances along the fiber with virtually negligible loss in intensity, even if the fiber is heavily bent. (A) State the two essential optical conditions that must be satisfied for total internal reflection to occur. (1 Mark) (B) Write the fundamental mathematical relation linking the critical angle (i_c) to the relative refractive index of the denser medium with respect to the rarer medium. (1 Mark) (C) If an optical fiber is manufactured such that the absolute refractive index of its core is **1.50** and that of its cladding is **1.40**, calculate the exact critical angle for the core-cladding interface. (Express the answer in terms of sine inverse). (2 Marks) [Chapter 9: Ray Optics | Confidence: Very High]

Section E: 5-Mark Questions (Long Answer)

Q31. (A) State Gauss's law in electrostatics. (B) Using Gauss's law, systematically derive the mathematical expression for the electric field intensity (E) due to a uniformly charged, very thin spherical shell of radius R and total charge Q , at a point: (i) strictly outside the shell ($r > R$). (ii) strictly inside the shell ($r < R$). (C) Plot a clearly labelled graph illustrating the variation of the electric field magnitude E as a function of the distance r from the geometric centre of the spherical shell. [Chapter 1: Electric Charges & Fields | Confidence: Very High]

Q32. (A) State the underlying principle and working of an electrical transformer. (B) Using Faraday's laws of induction, derive the mathematical relation between the primary and secondary voltages (V_p, V_s) and their respective number of turns (N_p, N_s) for an ideal transformer. (C) Why is the soft iron core of a transformer deliberately laminated in practical designs? (D) A step-down transformer is designed to reduce the main line voltage from **220 V** down to **11 V**. If the primary coil draws a steady current of **5 A** and the overall efficiency of the transformer is exactly **90%**, calculate the precise current delivered by the secondary coil. [Chapter 7: Alternating Current | Confidence: Very High]

Q33. (A) Draw a neat ray diagram showing the refraction of a monochromatic light ray at a convex spherical refracting surface separating two transparent media of refractive indices n_1 and n_2 ($n_2 > n_1$), when a point object is placed on the principal axis in the rarer medium. (B) Based on

this diagram, rigorously derive the standard relation: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$. (C) By successively applying this derived refraction formula to the two spherical surfaces of a thin convex lens, systematically deduce the Lens Maker's Formula: $\frac{1}{f} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$. [Chapter 9: Ray Optics | Confidence: Very High]

Answer Key

Q1: (b) 5. Charge $Q = ne \implies n = 8 \times 10^{-19} / 1.6 \times 10^{-19} = 5$ electrons. **Q2:** (c) Exactly zero. The entire equatorial plane of a dipole is an equipotential surface with $V = 0$. Work done $W = q_0(\Delta V) = 0$. **Q3:** (c) $v_d \propto E$. Drift velocity $v_d = (e\tau/m)E$, indicating a linear direct proportionality. **Q4:** (c) $4B$. $B = \mu_0 I / 2r$. New $B' = \mu_0(2I) / 2(r/2) = 4(\mu_0 I / 2r) = 4B$. **Q5:** (b) Paramagnetic material. Due to thermal agitation destroying the alignment of magnetic domains above the Curie point. **Q6:** (c) Energy. The mechanical work done in moving a magnet against the induced opposing force manifests as electrical energy. **Q7:** (c) 0. Phase angle $\phi = 90^\circ$. Power factor = $\cos(90^\circ) = 0$. **Q8:** (d) Gamma rays. They have the highest frequency and shortest wavelength, granting them immense penetrating ability. **Q9:** (b) $P/2$. Cutting a biconvex lens vertically makes each half a plano-convex lens. Its focal length doubles ($f' = 2f$), so its optical power is exactly halved ($P' = P/2$). **Q10:** (d) White. At the central maximum, the path difference for all constituent wavelengths is zero, meaning all colors interfere constructively to reform white light. **Q11:** (c) Remain entirely unchanged. Stopping potential depends solely on the frequency of incident light and the nature of the metal, entirely independent of macroscopic intensity. **Q12:** (b) A net torque but strictly zero net force. In a uniform field, $+qE$ and $-qE$ strictly cancel, leaving zero net force, but they form a couple that yields a torque.

Q13: (a) Both A and R are true and R is the correct explanation of A. Since $\Delta V = 0$, $W = \vec{F} \cdot d\vec{r} = q\vec{E} \cdot d\vec{r} = 0$, requiring $\vec{E}$ to be strictly perpendicular to $d\vec{r}$. **Q14:** (a) Both A and R are true and R is the correct explanation of A. **Q15:** (a) Both A and R are true and R is the correct explanation of A. The motional emf requires cutting magnetic flux ($e = Blv$); at the magnetic equator, the vertical field component is 0, so no flux lines are horizontally "cut" by the satellite's metallic body. **Q16:** (d) A is false and R is also false. KE strictly depends on frequency, not intensity. Intensity increases the number of photons, not their individual energies.

Q17: Original $C = \frac{\epsilon_0 A}{d} = 8 \text{ pF}$. New $C' = \frac{K\epsilon_0 A}{d/2} = 2K \left(\frac{\epsilon_0 A}{d} \right) = 2 \times 5 \times 8 = 80 \text{ pF}$. **Q18:**

Relation: $V = E - Ir$. When strictly short-circuited, $R_{ext} = 0$, so

$V = 0 \implies E = Ir \implies I = E/r$. $I = 1.5/0.5 = 3.0 \text{ A}$. **Q19:**

$B = \frac{\mu_0 NI}{2r} = \frac{(4\pi \times 10^{-7}) \times 50 \times 2.0}{2 \times 0.10} = \frac{400\pi \times 10^{-7}}{0.20} = 2000\pi \times 10^{-7} = 2\pi \times 10^{-4} \text{ T} \approx 6.28 \times 10^{-4} \text{ T}$

. **Q20:** (i) X-rays: Used in medical radiography (detecting fractures) and crystallography. (ii) Gamma rays: Used in targeted radiotherapy to destroy cancer cells and sterilizing medical equipment. **Q21:** Energy of incident photon $E = hc/\lambda \approx 1240/500 = 2.48 \text{ eV}$. Since $E < \Phi$ ($2.48 \text{ eV} < 3.0 \text{ eV}$), photoelectric emission will strictly **NOT** take place.

Q22: Under electric field E , force on electron $F = -eE$. Acceleration $a = -eE/m$. Drift velocity $v_d = a\tau = -(eE\tau)/m$. Magnitude $v_d = eE\tau/m$. Current

$I = neAv_d = neA(eE\tau/m) = \frac{ne^2 A\tau}{m} E$. Current density $J = I/A = \left(\frac{ne^2 \tau}{m} \right) E$. Hence,

$J = \sigma E$, where conductivity $\sigma = \frac{ne^2\tau}{m}$. **Q23:** Draw a long solenoid. Choose a rectangular Amperian loop ABCD of length L parallel to the axis, with arm AB completely inside and CD completely outside. $\oint \vec{B} \cdot d\vec{l} = \int_{AB} B dl + 0 + 0 + 0 = BL$. Enclosed current

$I_{enc} = (\text{number of turns in } L) \times I = nLI$. By Ampere's Law:

$BL = \mu_0(nLI) \Rightarrow B = \mu_0 nI$. **Q24:** Magnetic flux $\Phi = NBA \cos(\omega t)$. By Faraday's Law, induced emf

$\varepsilon = -d\Phi/dt = -NBA \frac{d}{dt} [\cos(\omega t)] = -NBA[-\omega \sin(\omega t)] = NBA\omega \sin(\omega t) = \varepsilon_0 \sin(\omega t)$.

Q25: Resonance occurs when the current amplitude is maximum, which strictly happens when inductive reactance equals capacitive reactance ($X_L = X_C$).

$\omega_r L = 1/(\omega_r C) \Rightarrow \omega_r^2 = 1/LC \Rightarrow f_r = \frac{1}{2\pi\sqrt{LC}}$. At resonance, total impedance $Z = R$,

so the phase angle $\phi = 0^\circ$. Power factor $\cos \phi = \cos(0^\circ) = 1$. **Q26:** Final image at infinity means intermediate image forms at the focus of the eyepiece ($v_e = f_e = 5 \text{ cm}$). Object distance for eyepiece is $u_e = -5 \text{ cm}$. Tube length $L = v_o + |u_e| \Rightarrow 15 = v_o + 5 \Rightarrow v_o = 10 \text{ cm}$. For the objective lens:

$1/v_o - 1/u_o = 1/f_o \Rightarrow 1/10 - 1/u_o = 1/1 \Rightarrow 1/u_o = 1/10 - 1 = -9/10$. Thus,

$u_o = -10/9 = -1.11 \text{ cm}$. **Q27:** Position of n^{th} bright fringe $y_n = n\lambda D/d$. For $n = 4$:

$$1.2 \times 10^{-2} = \frac{4 \times \lambda \times 1.0}{0.2 \times 10^{-3}}.$$

$\lambda = \frac{1.2 \times 10^{-2} \times 0.2 \times 10^{-3}}{4} = \frac{0.24 \times 10^{-5}}{4} = 0.06 \times 10^{-5} \text{ m} = 6 \times 10^{-7} \text{ m} = 600 \text{ nm}$. **Q28:** (A) Draw $+q$ and $-q$. Show curved equipotential surfaces around them, closely spaced in the region strictly between the charges, and a straight vertical line passing exactly through the midpoint. (B) For any point P on the equatorial plane at distance r from the centre, distance to both $+q$ and $-q$ is $x = \sqrt{r^2 + a^2}$. Total potential $V = V_+ + V_- = \frac{kq}{x} + \frac{k(-q)}{x} = 0$.

Q29: (A) Water (H_2O), or Hydrochloric acid (HCl). (B) The capacitance becomes perfectly infinite ($C \rightarrow \infty$) because a conductor has effectively $K = \infty$ (or E_{net} becomes zero). (C) Initial energy $U_0 = \frac{1}{2} C_0 V_0^2$. With the battery remaining connected, V stays exactly V_0 . New capacitance $C = KC_0 = 4C_0$. New energy $U = \frac{1}{2} C V_0^2 = \frac{1}{2} (4C_0) V_0^2 = 4U_0$. The energy increases exactly four times because the battery does extra work to pump more charge onto the plates. **Q30:** (A) 1. Light must exclusively travel from an optically denser medium to a rarer medium. 2. The angle of incidence must be strictly greater than the critical angle ($i > i_c$). (B) By Snell's Law at the boundary: $\sin i_c = n_r/n_d$. (C) $\sin i_c = n_{cladding}/n_{core} = 1.40/1.50 = 14/15$. Therefore, $i_c = \sin^{-1}(14/15)$.

Q31: (A) Total electric flux through any closed Gaussian surface is $1/\epsilon_0$ times the total net charge enclosed by the surface. (B) (i) For $r > R$, take a concentric spherical Gaussian surface of radius r . $\oint \vec{E} \cdot d\vec{S} = E(4\pi r^2) = Q/\epsilon_0 \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$. (ii) For $r < R$, the Gaussian surface encloses exactly zero charge. $E(4\pi r^2) = 0 \Rightarrow E = 0$. (C) Plot $E = 0$ from $r = 0$ up to $r = R$. At $r = R$, E spikes to a strict maximum $\frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$ and then rapidly decays as $1/r^2$ for $r > R$. **Q32:** (A)

Principle: Mutual Induction. A varying AC current in the primary coil generates a varying magnetic flux, which links with the secondary coil, inducing an emf. (B) $\varepsilon_p = -N_p(d\Phi/dt)$ and $\varepsilon_s = -N_s(d\Phi/dt)$. Assuming zero flux leakage, dividing them yields $V_s/V_p = N_s/N_p$. (C) To heavily strictly suppress eddy currents, thereby minimizing heat (energy) loss in the core. (D) Input Power $P_{in} = V_p I_p = 220 \times 5 = 1100 \text{ W}$. Output Power

$P_{out} = 90\% \text{ of } 1100 = 0.90 \times 1100 = 990 \text{ W}$. Also

$P_{out} = V_s I_s \Rightarrow 11 \times I_s = 990 \Rightarrow I_s = 90 \text{ A}$. **Q33:** (A) Draw convex surface separating

n_1, n_2 . Point object O in n_1 , refracted ray bends towards the normal to form real image I in n_2 . Mark centre of curvature C and angles i, r . (B) By Snell's law for small angles, $n_1 i = n_2 r$. From triangle geometry, exterior angle $i = \alpha + \gamma$ and $\gamma = r + \beta \implies r = \gamma - \beta$. Substitute to get $n_1(\alpha + \gamma) = n_2(\gamma - \beta)$. Using $\tan \theta \approx \theta$, substitute $\alpha \approx h/u, \beta \approx h/v, \gamma \approx h/R$ (applying proper sign conventions $u \rightarrow -u$). Result: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$. (C) Apply to Surface 1 (lens into air): $\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$. Apply to Surface 2 (lens out to air): $\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{R_2} = \frac{-(n_2 - n_1)}{R_2}$. Adding both explicitly yields $n_1(\frac{1}{v} - \frac{1}{u}) = (n_2 - n_1)(\frac{1}{R_1} - \frac{1}{R_2})$. Divide by n_1 and substitute $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ to obtain the Lens Maker's Formula.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.